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[4657]-573

S.E. (Computer Engineering) (I Semester) EXAMINATION, 2014

DIGITAL ELECTRONICS AND LOGIC DESIGN

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6 and Q. No. 7 or Q. No. 8.

(ii) Figures to the right indicate full marks.

(iii) Assume suitable data, if necessary.

1. (a) Do the following conversions : [6]

(i) $(101110.0101)_2 \rightarrow ()_{10}$

(ii) $(432A)_{16} \rightarrow ()_2$

(iii) $(428.10)_{10} \rightarrow ()_2$

P.T.O.

- (b) Reduce the following using K-map techniques : [4]

$$f(A, B, C, D) = \Pi(0, 2, 3, 8, 9, 12, 13, 15).$$

- (c) What is logic family ? Give the classification of logic family. [2]

Or

2. (a) Minimize the following expression using Quine-McClusky : [6]

$$f(A, B, C, D) = \Sigma m (0, 2, 3, 6, 7, 8, 10, 12, 13).$$

- (b) Explain with neat diagram two input CMOS NAND gate. [6]

3. (a) Explain Look Ahead Carry generator in detail. [6]

- (b) Explain with neat diagram working of serial- in serial-out 4-bit shift register. Draw necessary timing diagram. [6]

Or

4. (a) Explain rules for BCD addition with suitable example and design one digit BCD adder using IC 7483. [6]

- (b) Design given sequence generator using J-K FF. Sequence is
 $1 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 1$ [6]

5. (a) Draw the ASM chart for the following state machine. A 2-bit up counter is to be designed with output $Q_A Q_B$, and enable signal 'X'. If $X = 0$, then counter changes the state as $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$. If 'X' = 1, then counter should remain in current state. Design the circuit using JK-FF and suitable MUX. [7]
- (b) Write VHDL code for 4-bit adder using structural modeling style. [6]

Or

6. (a) Write a VHDL code for 8 : 1 MUX using Behavioural modeling. [7]
- (b) Draw an ASM chart, state diagram and state table for synchronous circuit having the following description.

The circuit has control input C, clock and outputs x , y , and z .

- (i) If $C = 1$, on every clock rising edge the code on output x , y and z changes from $000 \rightarrow 010 \rightarrow 100 \rightarrow 110 \rightarrow 000$ and repeats.

(ii) If $C = 0$, the circuit holds the present state. [6]

7. (a) Draw and explain Basic architecture of FPGA in detail. [6]

(b) Implement the following functions using PLA : [7]

$$f_1(A, B, C) = \Sigma m(0, 3, 4, 7)$$

$$f_2(A, B, C) = \Sigma m(1, 2, 5, 7).$$

Or

8. (a) A combinational circuit is defined by the functions :

$$f_1(A, B, C) = \Sigma m(3, 5, 7)$$

$$f_2(A, B, C) = \Sigma m(4, 5, 7).$$

Implement the circuit with PLA having 3 input and 3 product term with 2 output. [7]

(b) Implement 4 : 1 MUX using PAL. [6]